

NATS CLI Cheatsheet

Overview

Download: <https://github.com/nats-io/natscli>

One stop command line utility to interact with and manage NATS.

Independent from NATS server version. Uses only public Golang API.

Please update regularly or compile the GitHub main branch.

Getting started

Built-in cheat sheet

nats cheat

Always check help for additional options

nats server info --help

Quick system status

nats server list

Show raw nats cli messages. No secret sauce

nats server list --trace

Publish/Subscribe

Listen to a subject

nats sub foo

Publish to a subject

nats pub foo Hello

Listen to wildcards

nats sub '>'

nats sub 'orders.emea.*'

nats sub 'orders.*.01234'

Context and credentials

List local connection contexts

nats context list

Store credentials and server URL for *cloud*

nats context add --server

nats://mycloud.io --user

admin --password admin cloud

Select context

nats context select cloud

Request/Reply

Simple reply mockup

nats reply bar World

Send a request

nats request bar Hello

JetStream

List streams, including offline and inaccessible streams

nats stream list

Create stream interactively and with defaults

nats stream add <stream>

nats stream add <stream>

--replicas 3 --defaults

Stream view content

nats stream view <stream>

Publish JetStream message with ack

nats pub --jetstream foo Hello

Key-Value Store

List Key-Value buckets

nats kv list

Create a KV bucket

nats kv add <bucket>

Put and get a key

nats kv put <bucket> foo Hello

nats kv get <bucket> foo

Watch for changes

nats kv watch <bucket> '>'

Distributed security

Create an operator

nats auth operator add

my-system

Generate a sample config

nats server generate

server.conf

Walkthrough with accounts and users

nats auth cheat

Server Health and Monitoring

Show system health (SYS)

nats server list

Detailed server information (SYS)

nats server info <server>

Check account traffic

nats traffic

Gather and analyze for debugging (SYS)

nats audit gather

nats audit analyze <audit zip>

JetStream Monitoring

Show JetStream and meta status (SYS)

nats server report jetstream

--watch=1

List all streams

nats stream report

Stream details

nats stream info <stream>

JetStream backup and purge

Backup a stream with consumer state

nats stream backup <stream> <dir>

Restore stream from backup

nats stream restore <file>

Backup just messages

nats stream backup --no-consumers

<stream> <dir>

Backup all streams in current account

nats account backup <dir>

Purge a stream

nats stream purge <stream>

Purge a subject in a stream

nats stream purge --subject foo

<stream>

Trace and debug message flow

Trace message without delivering to clients

nats trace foo Hello

Trace with delivery and timestamps

nats trace --timestamp --deliver foo

Measure roundtrip and latency

Roundtrip single client to server

nats --server srv1:4222 rtt

Measure latency between clients connected to different servers

nats latency --server srv1:4222

--server-b srv2:4222

Benchmarking

Subscribe core NATS for 5 clients

nats bench sub foo --clients 5

--msgs 10000

Publish core NATS with 10 publishers

nats bench pub foo --clients 10

--msgs 10000 --size 512

Request-reply server for core nats

nats bench service serve --clients 5

testservice

Benchmark core nats subscribe for 5 clients

nats bench service request --clients

5 testservice --msgs 10000

JetStream async acknowledged publishing of batches of 100

nats bench js pub foo --create

--batch 100

JetStream sync publishing using 10 clients - purge stream

nats bench js pub foo --purge

--batch=1 --clients=10 --msgs 10000

JetStream delivery from a stream through a durable consumer with 4 instances

nats bench js consume --clients 4

--msgs 10000

Dashboards - top style

Watch all servers (SYS)

nats server watch servers

Watch a single server with statistics (SYS)

nats top <server-name>